Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

# DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Electromagnetic spectrum

Standalone video lesson · authored study notes with source citations

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# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

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M1

# Electromagnetic spectrum

## 1 Electromagnetic spectrum

Study focus: Electromagnetic spectrum. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

**Core explanation · 00:00–04:59**

We will continue with module one on molecular spectroscopy and in today's class we are going to look at electromagnetic spectrum and the information obtained from each region of the spectrum.

**Key details**

- Light is a form of electromagnetic radiation, which is divided into different regions based on the frequency of the radiation.
- Though these demarcations are made, the boundaries are not very precise.
- Then we have infrared radiation, then we have the visible region, then we go on to ultraviolet region.
- Then more energetic X rays and finally the gamma rays which are high energetic.
- First, the lowest energy beam, that is the radio frequency region.
- The wavelength is ten meters to one centimeter.
- That clearly conveys that the energy of this radiation is very low.
- It is known as NMR and ESR.
- NMR stands for nuclear magnetic resonance and ESR stands for electron spin resonance.
- Electron spin resonance is also referred to as electron paramagnetic resonance sometimes.
- What exactly do happens when radio frequency region light falls onto the sample.
- In the case of NMR that is called that is nuclear magnetic resonance there's going to be nuclear spin reversal, whereas in the case of ESR there's going to be electron spin reversal.
- We will start with electron spin because that is something you already understand that we can have two spins that is plus half and minus half.
- And both of them occupy the same orbital.
- So that means they are degenerate.
- Both these spins, plus and minus half, are degenerate and they occupy the same orbital.

- But in the presence of magnetic field, what happens is these will split into two levels one will be the lower level that is say minus half and the other one will be plus half.
- What is the difference?
- Say in one case it is the spin is clockwise and in the other case the spin is anti-clockwise.
- When you have a magnetic field, when these electrons are placed in a magnetic field, we find that while the energy levels now split into one with lower energy and one with higher energy, it is basically.
- How did this happen?
- Because the electron is spinning and it interacts with the magnetic field.
- One could be aligned with the magnetic field, the other could be opposite in direction to the magnetic field.
- Therefore, we get two energy levels now.

**Example or application from the lecture**

- Let us see what is the information we obtain from the different regions of the electromagnetic spectrum.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

**Core explanation · 05:00–09:59**

When a radiation falls on this atom, then we will find that if the energy gap is equivalent to the frequency of the radiation which is falling on this atom.

**Key details**

- then there will be a transition of the atom from one energy level to the next energy level.
- It is going to absorb this radiation and move from one level to the ne to from the lower level to the higher energy level.
- And what has happened because of this movement?
- There is a spin reversal.
- And why are we calling it as a resonance phenomenon?
- The energy gap.
- The energy gap means the frequency matches with the frequency of the radiation we know resonance will happen and the atom now picks up this energy and goes from the lower energy level to the higher energy level.
- But when there is a magnetic field present, then there is splitting, just like it happened in the electron.
- For the electron, the same phenomena happens for nuclei also.
- The spins are now you have one where it is aligned with the magnetic field and the other one where it is opposite to the magnetic field.
- And now we have two energy levels.
- When radiation is made to fall on these nuclei, then they are going whenever the energy gap matches with the incident.

- light frequency, incident radiation frequency, the atom now the nuclei are going to absorb that energy.
- There will be a resonance phenomenon.
- The nuclei will absorb energy and go to the next level.
- Again what happened?
- What we observe here is nuclear spin reversal.
- In the case of nuclei this would be plus my plus half and this would be minus half.
- that the energy gap is not very large.
- Therefore, a low frequency beam like the radio frequency region is sufficient to cause the spin reversal of either the nuclei or the electron based on what is the spectroscopy we are studying.
- Now here the wavelength is 1 centimeter to 100 micrometers means it is slightly more energetic beam and the spectroscopy studied in this region is rotational spectroscopy and we talk of rotational levels.
- We come to know of rotational levels.
- When the molecule rotates.
- There is a change in the dipole moment.
- We know that XL is a polar molecule.
- There is a change in the dipole moment.
- So as the molecule rotates, there is bound to be the fluctuation in the electric field because of the change in dipole moment.
- with which the electric field is fluctuating.
- That also has a fluctuating electric field because it's an electromagnetic radiation, there's a fluctuating electric field and a fluctuating magnetic field.
- Fluctuating electric field of the molecule due to rotation matches with the fluctuating electric field of the radiation falling on the molecule, resonance phenomena will happen, and the molecule now will move from.

**Example or application from the lecture**

- Now let us come to the next region that is the microwave region.

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EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

**Core explanation · 10:00–14:59**

one rotational level to the next rotational level.

**Key details**

- Because it is by virtue of the rotation of the molecule.
- If we have homonuclear molecules like say hydrogen, then we find that we cannot study rotational spectroscopy.
- Because even when the molecule rotates, there is no rotating electric field, and if you drop radiation.

- the incident radiation onto this molecule, we will find that there will be no interaction as there is no fluctuating electric field.
- And the study we do here is known as vibrational spectroscopy, and now we will learn about vibrational levels.
- When a molecule undergoes vibration, say a diatomic molecule, simple diatomic molecule, when it undergoes vibration, again, say it is stretching.
- The frequency of the elect fluctuating electric field due to vibration of the molecule, the molecule is going to absorb this radiation, there will be a resonance like phenomena.
- Molecule will absorb radiation and it will move to the next level.
- We call it called as the vibrational levels.
- Because it is due to the vibrational motion of the molecule.
- One thing we can see very clearly here that rotational transition requires the.
- Radiation to be in the microwave region and whereas for vibrational transitions the we require infrared region which is much more energetic.
- Therefore, it requires infrared region, whereas in the microwave region they are closer as it the transition requires only the less energetic beam as compared to infrared.
- It's only when you come to triatomic molecules you can notice the bending motion also.
- The dipole moment, hence no change in electric field.
- So therefore no fluctuating electric field and there will be no interaction with the incident radiation.
- And therefore, we say that the homonuclear diatomic molecules again are not active in the infrared region.
- We know visible comes first and then the UV region, but nevertheless the spectrum with the spectroscopy is referred to as UV visible spectroscopy and the wavelength now has become one micrometer to ten nanometers.
- Here now we are going to look at electronic states.
- Because we understand that the molecule or the atom is going to absorb energy and.

**Example or application from the lecture**

- Let us get to the next beam of next region that is the visible and the UV region.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

**Core explanation · 15:00–19:59**

the m atom will go to the from the electron now transits from one atomic level to the next atomic level.

**Key details**

- Similarly in a molecule it goes from one molecular orbital to the next molecular orbital.
- There are electronic transitions happening and that will happen between the different electronic states.
- We must understand it's only from the Homo to the lumon, that is the highest occupied molecular orbitals in the case of molecules.

- That will be the valence electrons, the far the higher energy electr electrons because they require the minimum energy t the energy is required to move them is much less as compared to the one the an electrons which are closer to the nuclei as in the case of atoms and in this case in the molecule also it will move from Homo to lumo.
- Therefore there is change in electronic disposal.
- Distribution of the valence electrons.
- U if it happens in the visible region we can see the color.
- If it's in the UV region, obviously we cannot see the colour.
- These are very energetic beams and now the wavelength has come down to picometers, so 10 nanometers to 100 picometers, and this study is known as X-ray spectroscopy.
- But these transitions now will be involving the inner electrons.
- In the case of UV visible, it is the valence electrons which undergo transition and that requires much less energy.
- Whereas if you want to remove the inner electrons then it requires more energy and therefore x-ray region is what is required the light in the X ray region, the frequency should be high, energy should be more to cause this electronic distribution which involves the inner electrons.
- We are aware that gamma rays are produced whenever there is nuclear reactions happening.
- We know that nuclei consists of protons and neutrons which are very tightly held.
- we have to use very highly energetic beams in order to cause any nuclear rearrangement.

**Example or application from the lecture**

- We know that whenever the radioactive nuclei changes into another nuclei, gamma rays are given out.
- In all the other cases we are concentrating more on the absorption spectroscopy, but here we will study more it's the emission spectroscopy and based on the gamma rate which is being given out, we can get information about what in nuclear rearrangement happened and what were the nuclei which are getting formed or what is the nuclear reaction which is happening.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

**Core explanation · 20:00–24:59**

is giving us different information about the atom or the molecule which is being impinged upon by the radiation.

**Key details**

- One more type of spectroscopy which is very important is Raman spectroscopy.
- But it is different from all these like rotational or vibrational spectroscopy.
- or electronic spectroscopy.
- Because there we talk about absorption of light, there or emission of light.
- But here Raman spectroscopy talks about scattering of light.
- Radiation will get transmitted.
- Most of the radiation we expect apart from what is absorbed is going to be transmitted.

- transmitted radiation, but some amount of radiation there will be a change in direction and this is referred to as the scattered radiation.
- Normally in transparent medium we expect very little scattering, but nevertheless there is a possibility that scattering will happen.
- Whatever incident radiation is there, the same wavelength will be seen most of the time for the scattered radiation and such a scattering is referred to as.
- Ra r rarely scattering.
- But Rama noticed that some of the scattered radiation consists of radiation of a different wavelength from that of the incident wavelength.
- Most of the times the scattering which happens is going to be rally scattering where the incident wavelength and the scattered the incident light wavelength and the scattered light wavelength has the has is same.
- There will be some radiations which we find that the wavelength is more.
- of the scattered radiation which we refer to as the Stokes Raman scattering or they when you see in the spectrum they are called as Stokes lines and some of the radiation we will find will have a smaller wavelength and those are referred to as the anti scale.
- Stokes, Raman, scattering, and in the spectrum they are referred to as the anti-stokes lines.
- And that these photons carry.
- E is equal to H nu.
- We know that E is equal to H nu.
- And again it cannot remain there because it's an unstable state from the ground state.
- From the ground state it goes to a virtual state and then it comes back with the same frequency of radiation which is referred to as elastic scattering.
- same kind the same frequency of radiation and this is referred to as valley scattering or this is elastic in nature, elastic scattering.
- But there could be what Raman observed, but there were some radiations which were inelastic scatter were inelastic.
- scattering was happening.

**Example or application from the lecture**

- Let us see what are these Stokes and Antistokes like now we can e we can explain this quantum mechanically if you consider light to be made up of photons.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

**Core explanation · 25:00–29:59**

initial ground state it ends up at a higher level and therefore now the frequency is much less.

**Key details**

- See to a higher level, therefore the energy of this beam is much less.
- It is now you have to subtract this much of energy.
- It is not at the ground level, it is starting at a higher level, it goes up.
- But when it comes back, then there the it comes back to the ground state.

- That means now the energy is more.
- You can see the energy gap, it is much more and therefore now the beam has more energy or smaller wavelength, and this is known as anti stokes scattering.
- And we find that.
- Whenever this scattering happens, inelastic scattering, there could be you will see stokes lines as well as anti-stokes lines.
- Of course, it is found that the stokes lines are much more intense as compared to the anti-stokes lines.
- That is because the population of the ground state will be any time more than that of the higher level.
- So anti-stokes lines will appear only if there are molecules in this higher level they go absorb energy and then come back to the initial ground state and therefore their energy they are much more.
- And one advantage of Raman spectroscopy is that homonuclear diatomic molecules I mentioned just now will not show any absorption in the microwave or the.
- all the information which we couldn't get from the pure rotational or pure vibrational spectroscopy we can obtain from Raman spectroscopy.
- And of course we can see as we go from one region to the other, the frequency is increasing, the wave number is decreasing.
- Length is decreasing and the wave number is increasing.
- So in the next class we are going to talk about Born Oppenheimer approximation and then we will also move to rotational spectroscopy.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Electromagnetic spectrum transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| Electromagnetic spectrum transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Electromagnetic spectrum transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps. |
| Electromagnetic spectrum transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

## Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
| --- | --- | --- |
| SRC00006 | 6 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare... |
| SRC00007 | 6 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00008 | 6 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare... |
| SRC00009 | 6 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare... |

# C Revision and retrieval practice